



DAY 01: What's Artificial Intelligence & it's Applications

What is Artificial Intelligence?



A Computer Science Perspective

AI is the study and development of computational systems that can perform tasks requiring human intelligence through algorithms, data structures, and mathematical models.



Algorithms

Search, optimization,
learning



Data Structures

Trees, graphs, vectors



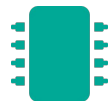
Mathematics

Linear algebra, calculus,
stats

What is Artificial Intelligence?



Artificial Intelligence (AI) is the simulation of human intelligence processes by machines, especially computer systems. These processes include learning, reasoning, and self-correction.



Machine Learning

Systems that learn from data



Problem Solving

Automated decision-making



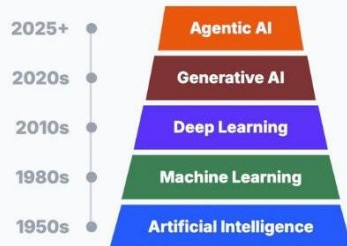
Pattern Recognition

Identifying trends & insights

The AI Timeline

Evolution of the AI stack from the 1950s to today.

The AI Stack



The New Frontier

2025 - Future

Agentic AI

Autonomous systems that plan and execute complex workflows using digital tools. This technology shifts the focus from simple chat interactions to independent enterprise action.

Adoption Forecast (Capgemini)

37%

37% of enterprises are now piloting or scaling AI agents as tools transition from assistants to orchestration systems.

The Innovation

2020s

Generative AI

Advanced models that synthesize data to create original text, code, and images from human prompts. This is the era of creative collaboration between humans and machines.

The Scale

2010s

Deep Learning

The neural layer. Using layered networks to master perception, allowing machines to finally recognize speech and vision with high human precision across digital datasets.

The Shift

1980s

Machine Learning

The statistical engine. Moving from manual rules to systems that learn from examples to identify complex patterns and predict future outcomes with very high accuracy.

The Origin

1950s

Artificial Intelligence

The foundational discipline. A field focused on building systems capable of logic and reasoning to solve problems originally handled by highly trained human experts.

AI Vocabulary

Every team must understand (2026)



1. Artificial Intelligence

Tech that lets machines perform tasks using human-like reasoning.



2. Machine Learning

AI that studies data patterns and improves its output automatically.



3. Deep Learning

ML that uses layered networks to solve advanced and complex problems.



4. Neural Network

A linked system of nodes that learns patterns from large datasets.



5. Generative AI

AI that produces new text, images, audio, or video from learned data.



6. Large Language Model

A model trained on huge text sets to understand and generate language.



7. AI Agent

A system that completes tasks and makes decisions without supervision.



8. Agentic AI

AI that plans, reasons, and executes actions through independent steps.



9. Chatbot

AI software that holds natural language conversations with users.



10. Prompt

A written instruction that tells an AI system what response to create.



11. Fine Tuning

Extra training that adapts an AI model to a specific, narrow purpose.



12. Data Training

Feeding labeled data into AI so it learns patterns and correct behavior.



13. Dataset

A structured collection of information used to train or evaluate models.



14. Model

A trained system that turns input into predictions or generated output.



15. Automation

AI handling repetitive work so tasks run smoothly without manual input.



16. Computer Vision

AI that interprets images or video to detect and understand visual content.



17. Natural Language Processing

AI that reads language, interprets meaning, and produces clear responses.



18. Voice AI

AI that recognizes speech, interprets intent, and returns spoken results.



19. Personalization

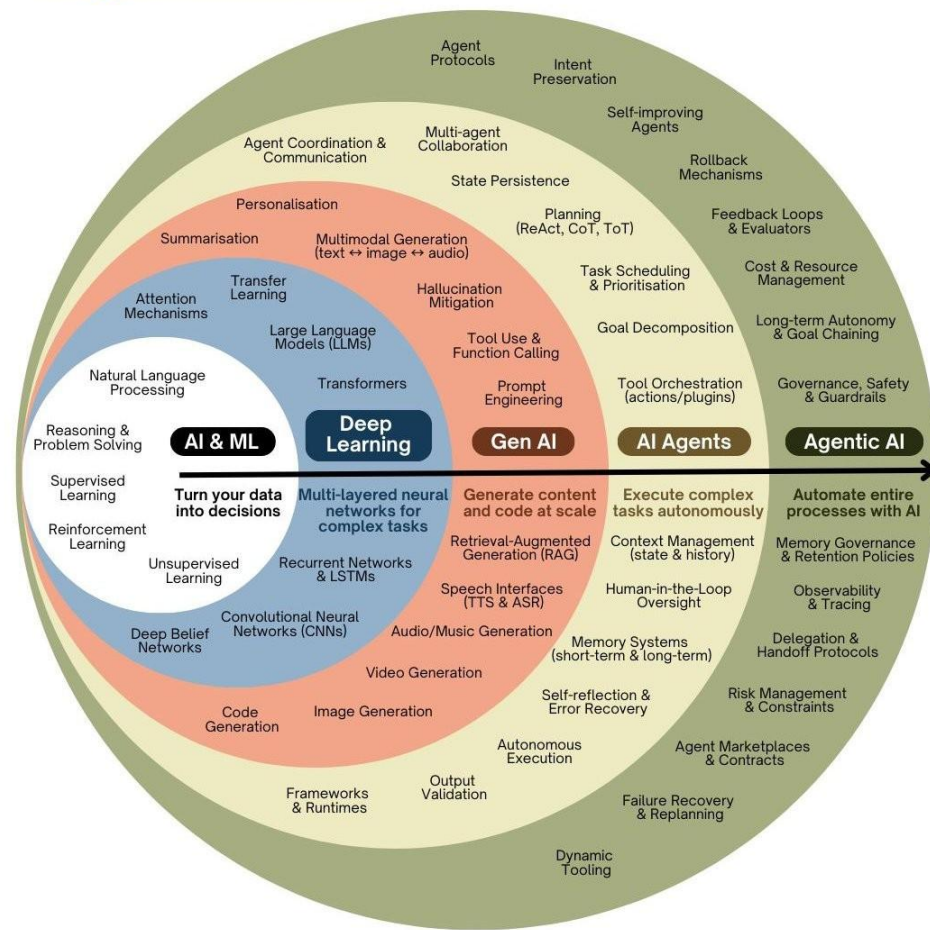
AI adjusting content or actions based on a user's data and preferences.



20. AI Ethics

Rules that guide AI use to stay fair, transparent, and socially safe.

Agentic AI: The Big Picture



AI Market & Possibilities

\$827B

Global AI Market Size (2030)

Key Application Areas

- Healthcare: Disease diagnosis, drug discovery, personalized medicine.
- Finance: Fraud detection, algorithmic trading, risk assessment.
- Transportation: Autonomous vehicles, traffic optimization, logistics.
- Retail: Recommendation systems, inventory management, customer service.
- Manufacturing: Quality control, predictive maintenance, supply chain optimization.

Nepali Language AI Applications

Nepali Chatbots

Examples: Customer service in Nepali • Government information bots • Healthcare assistance in local language

✓ *Accessibility for non-English speakers*

Document Processing

Examples: OCR for Devanagari script • Legal document analysis • Historic manuscript digitization

✓ *Preserve and digitize Nepali heritage*

Translation Services

Examples: Nepali ↔ English translation • Multi-dialect support • Real-time voice translation

✓ *Break language barriers*

Types of AI

Based on Capability

Narrow AI (ANI)

Does ONE task extremely well
Ex: Siri, Chess engines, spam filters

General AI (AGI)

Human-level thinking across
any domain — still theoretical

Super AI (ASI)

Surpasses human intelligence
in all areas — future concept

Based on Function

Reactive Machines

No memory, reacts to current input
Ex: IBM Deep Blue

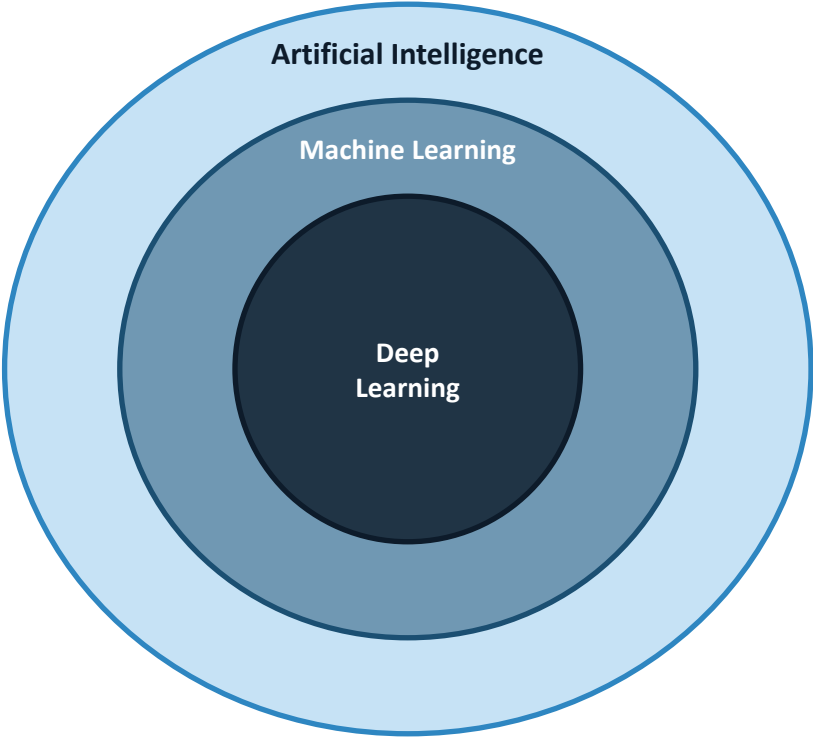
Limited Memory

Uses past data for decisions
Ex: Self-driving cars

Theory of Mind

Understands human emotions
& intent — in research

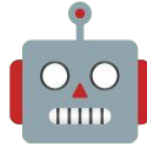
AI vs Machine Learning vs Deep Learning



	AI	ML	DL
Definition	Simulate human intelligence	Learns from data	Uses neural networks
Needs Data?	Not always	Yes	Large amounts
Example	Expert systems	Spam filters	Image recognition
Hardware	Standard CPU	Standard CPU	GPU / TPU needed

Machine Learning

The Three Main Paradigms



Supervised Learning

What is it?

Learning from labeled data

How it works:

1. Train with input-output pairs
2. Model learns the mapping
3. Predict outputs for new inputs

Like a teacher showing examples

Common Applications

✓ Email Spam Detection

Classify emails as spam/not spam

✓ House Price Prediction

Predict price from features

✓ Image Classification

Label images (cat, dog, car)

✓ Medical Diagnosis

Detect diseases from symptoms

✓ Credit Scoring

Approve/reject loan applications

Unsupervised Learning

What is it?

Learning from unlabeled data

How it works:

1. Find patterns in data
2. Group similar items together
3. Discover hidden structure

Like exploring without a map

Common Applications

- ✓ **Customer Segmentation**
Group customers by behavior
- ✓ **Anomaly Detection**
Find unusual transactions
- ✓ **Recommendation Systems**
Suggest similar products
- ✓ **Topic Modeling**
Discover themes in documents
- ✓ **Data Compression**
Reduce dimensions (PCA)

Reinforcement Learning

What is it?

Learning through trial and error

How it works:

1. Agent takes actions
2. Receives rewards/penalties
3. Learns optimal strategy

Like training a pet with treats

Common Applications

✓ Game AI

AlphaGo, chess engines

✓ Robotics

Robot navigation and control

✓ Self-Driving Cars

Autonomous vehicle decisions

✓ Resource Management

Data center optimization

✓ Trading Bots

Stock market strategies

Machine Learning Types Comparison

Type	Data Type	Goal	Example
Supervised	Labeled data	Predict output from input	Email spam filter
Unsupervised	Unlabeled data	Find hidden patterns	Customer segmentation
Reinforcement	Interaction data	Learn optimal actions	Game playing AI

Simple Analogy:

Supervised = Learning with a teacher giving correct answers • Unsupervised = Self-study discovering patterns • Reinforcement = Learning from experience (trial & error)

Deep Learning

Neural Networks with Multiple Layers

A subset of machine learning using artificial neural networks inspired by the human brain

Deep Learning: How It Works

1

Input Layer

Receives raw data (images, text, numbers)

Ex: Pixels of an image

2

Hidden Layers

Multiple layers extract features progressively

Ex: Edges → Shapes → Objects

3

Output Layer

Produces final prediction or classification

Ex: Cat, Dog, or Bird

Types of Deep Learning Models

Type	Full Name	Best For	Example
CNN	Convolutional Neural Networks	Image recognition, computer vision	Face detection, self-driving cars
RNN	Recurrent Neural Networks	Sequential data, time series	Language translation, speech recognition
LSTM	Long Short-Term Memory	Long sequences, memory retention	Text generation, stock prediction
GAN	Generative Adversarial Networks	Generate new data	Image synthesis, deepfakes
Transformer	Transformer Models	NLP, attention mechanisms	ChatGPT, BERT, translation

Generative AI

Creates original content from learned patterns

Text Generation

ChatGPT, Claude, Gemini

Writing, coding, translation

Image Generation

DALL-E, Midjourney, Stable Diffusion

Art, design, photography

Audio/Music

Suno, Udio, ElevenLabs

Music composition, voice cloning

Video Generation

Sora, Runway, Pika

Video creation, editing

Code Generation

GitHub Copilot, Cursor, Replit

Programming assistance

3D/Science

AlphaFold, molecular design

Protein folding, drug discovery

Real-World AI Applications



Healthcare

Disease detection, drug discovery, medical imaging.



Finance

Fraud detection, stock trading, credit scoring.



E-Commerce

Recommendation engines, chatbots, dynamic pricing.



Transport

Self-driving cars, route optimization, predictive maintenance.



Social Media

Content feed curation, face recognition, moderation.



Education

Personalized learning, automated grading, tutoring bots.

Industry Use Cases & Trends

\$1.8T

Global AI Market by 2030

New AI Jobs are been arising

83%

Companies see AI as priority



Generative AI:

ChatGPT, Gemini, DALL-E — creating text, images, code from prompts



AI in Edge Devices:

Running models on phones & IoT without cloud dependency



Explainable AI (XAI):

Making AI decisions transparent and understandable



AI + Robotics:

Humanoid robots and autonomous warehouses becoming real



Agentic AI:

AI that can autonomously plan, execute multi-step tasks

Challenges in Studying AI



Understanding the Complexities

Steep Learning Curve

Requires strong foundation in mathematics (calculus, linear algebra, statistics), programming, and algorithmic thinking.

Rapidly Evolving Field

New techniques, frameworks, and best practices emerge constantly, requiring continuous learning and adaptation.

Computational Resources

Training complex AI models requires significant computing power, which can be expensive and inaccessible for beginners.

Data Quality & Availability

Access to large, high-quality datasets is essential but often restricted due to privacy, cost, or licensing constraints.

Technical & Resource Challenges

Limited Computing Power

Issue: AI training requires expensive GPUs and cloud resources

✓ *Solution: Use cloud credits (AWS, Google), free tiers, Google Colab, Kaggle notebooks*

Insufficient Local Data

Issue: Lack of Nepali language datasets and labeled data

✓ *Solution: Contribute to open datasets, crowdsource data collection, synthetic data generation*

Infrastructure Gaps

Issue: Unreliable internet, power cuts, limited bandwidth

✓ *Solution: Offline learning resources, mobile hotspots, distributed learning schedules*

High Learning Curve

Issue: Math, programming, and domain knowledge required

✓ *Solution: Start with basics, use structured courses (Coursera, edX), join study groups*

Overcoming the Challenges



Strategies for Success

- Start with fundamentals: Master programming (Python), mathematics, and core ML concepts before diving into advanced topics.
- Hands-on practice: Work on projects, participate in Kaggle competitions, contribute to open-source AI projects.
- Leverage free resources: Use platforms like Coursera, edX, fast.ai, and Google Colab for learning and experimentation.
- Join communities: Engage with AI communities on GitHub, Reddit, Discord, and attend meetups or conferences.
- Stay updated: Follow research papers, blogs, and thought leaders; read publications like arXiv, Towards Data Science.
- Specialize gradually: After building a broad foundation, focus on specific domains like NLP, computer vision, or reinforcement learning.

Essential Learning Resources



Curated for Students



Online Courses

- Andrew Ng - ML (Coursera)
- Stanford CS229 - ML
- MIT 6.034 - AI
- fast.ai - Practical Deep Learning
- Berkeley CS188 - AI



Tools & Platforms

Python, NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, Jupyter



Recommended Books

- Hands-On ML (Géron)
- Deep Learning (Goodfellow)
- Pattern Recognition (Bishop)
- AI: Modern Approach (Russell)



Communities

Kaggle, Papers with Code, r/MachineLearning, Hugging Face, GitHub

Key Takeaways

- 1 AI is transforming every industry globally and in Nepal.
- 2 Nepal has unique opportunities in agriculture, tourism, banking.
- 3 Challenges exist but solutions are available - start learning today
- 4 Growing demand for AI professionals creates career opportunities
- 5 Free resources and communities support your learning journey

The Future is Now

AI + Human = Unlimited Potential

Start learning today.

Build solutions for Nepal.

Be part of the AI revolution.

Thank You!

